Maple: Contrastive Retrieval

1 Introduction

This document is a controlled fixture for evaluating PDF evidence retrieval. All datasets and experimental results below are fictional.

2 Method

Our method uses dense retrieval and a contrastive encoder. The system retains source coordinates for every passage.

3 Datasets

We evaluate on the fictional MapleQA dataset, containing 1800 training questions and 400 test questions.

4 Evaluation metrics

The evaluation metric is macro F1. The reported macro F1 score is 0.78 on MapleQA.

5 Ablation study

Removing contrastive training reduces macro F1 from 0.78 to 0.69. The ablation measures the impact of contrastive supervision.

6 Limitations

The method requires GPU training and was tested on one fictional dataset. Performance on long documents was not evaluated.

7 Notes

No hardware latency, power consumption, clinical outcome, or deployment cost was measured. This fixture is intended for software testing only.